TAPASH KUMAR

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SUMMARY

AI Engineer with experience building production-grade AI systems, LLM-powered applications, and scalable backend services. Skilled in designing agentic workflows, Retrieval-Augmented Generation (RAG) pipelines, semantic search, and cloud-native APIs using FastAPI, Node.js, and modern AI frameworks. Passionate about building reliable software that combines distributed systems with applied machine learning.

TECHNICAL SKILLS

- **Programming Languages:** Python, TypeScript, JavaScript (ES6+), SQL
- **Frontend Development:** Next.js, React.js, HTML5, CSS3, Tailwind CSS, Bootstrap
- **Backend Development:** Node.js, NestJS, Express.js, FastAPI, Flask, REST APIs, WebSockets, JWT Authentication
- **Generative AI & LLMs:** OpenAI API, Azure OpenAI, LangChain, LangGraph, Hugging Face Transformers, Prompt Engineering, AI Agents, Function Calling, Structured Outputs, Retrieval-Augmented Generation (RAG)
- **Vector Search & AI Retrieval:** FAISS, ChromaDB, Pinecone, Vector Embeddings, Semantic Search, Hybrid Search, Document Chunking, Metadata Filtering
- **Machine Learning & Deep Learning:** Scikit-learn, TensorFlow, Keras, XGBoost, CNNs, NLP, Model Training, Hyperparameter Tuning, Feature Engineering, Model Evaluation
- **Databases:** PostgreSQL, Microsoft SQL Server, SQLite, Prisma ORM
- **Cloud & DevOps:** Microsoft Azure, Docker, Git, GitHub, CI/CD, Linux, Firebase Authentication
- **Data Science & Visualization**: NumPy, Pandas, Matplotlib, Seaborn, Exploratory Data Analysis (EDA), Statistical Analysis, Data Cleaning
- **Tools & Platforms:** Postman, Jupyter Notebook, Streamlit

EXPERIENCE

AI Engineer - InfoZeb Consultancy, Gurugram — March 2026 – Present

- Architected and developed an AI-powered HR automation platform integrating recruitment, onboarding, employee management, and conversational AI into a unified workflow, reducing manual HR operations through intelligent automation.
- Designed and implemented production-grade Retrieval-Augmented Generation (RAG) pipelines using LangChain, vector embeddings, and semantic search to deliver accurate, context-aware responses for candidate screening, resume analysis, and HR knowledge retrieval.
- Engineered stateful multi-agent workflows with **LangGraph**, enabling autonomous decision-making, human-in-the-loop approvals, interview scheduling, and configurable business workflows through dynamic graph orchestration.
- Built scalable backend services and REST APIs using **FastAPI** and **TypeScript**-based microservices, integrating LLM inference, structured outputs, authentication, and enterprise HRMS modules while following modular, API-first architecture principles.
- Optimized LLM performance through advanced prompt engineering, structured output validation, retrieval optimization, and tool/function calling, improving response consistency, reducing hallucinations, and enhancing production reliability.
- Collaborated across product, frontend, and backend teams to deliver end-to-end AI features, integrating recruiter dashboards, conversational interfaces, and enterprise workflows while ensuring maintainable, scalable, and reusable AI infrastructure.

AI Engineering Intern - Stabor India Private Limited, Gurugram — December 2025 – February 2026

- Developed an end-to-end demand forecasting solution using XGBoost for real-world industrial datasets, designing robust data preprocessing pipelines, feature engineering strategies, and regression models to improve prediction accuracy by **15–20%**.
- Engineered time-series features including lag variables, rolling window statistics, and temporal attributes, enabling the model to capture seasonal trends, demand fluctuations, and long-term consumption patterns with improved generalization.
- Optimized model performance through extensive hyperparameter tuning, cross-validation, and L1/L2 regularization techniques, significantly reducing overfitting while improving prediction stability across unseen datasets.
- Built automated data processing and evaluation pipelines using Python, ensuring efficient handling of large datasets and validating model performance using industry-standard metrics such as RMSE, MAE, and R^2 for reliable forecasting.

Intern - CSIR – National Physical Laboratory (NPL), New Delhi — June 2025 – July 2025

- Developed computational workflows to analyze atomic clock frequency stability using Allan Deviation, processing large-scale time-series measurement data to characterize noise behavior and long-term system performance.
- Performed statistical analysis of oscillator noise, drift estimation, and frequency stability across multiple averaging intervals, benchmarking results against international **BIPM** standards to ensure measurement accuracy and reliability.
- Automated scientific data processing and visualization pipelines using Python, reducing manual analysis efforts while improving the efficiency and reproducibility of metrology experiments.
- Collaborated with researchers to interpret experimental results, validate measurement methodologies, and support precision timekeeping research through quantitative analysis and data-driven insights.

Intern - ISRO Telemetry, Tracking and Command Network (ISTRAC), Bangalore — June 2024 – July 2024

- Designed and optimized microwave cavity models using **Ansys HFSS** for Hydrogen Maser applications, achieving a resonant frequency within **~44 Hz** of the target **TE011** mode through iterative electromagnetic simulations.
- Conducted parametric studies and electromagnetic field analysis to optimize cavity geometry, material properties, and quality factor (Q-factor), improving resonance stability and overall system performance.
- Analyzed simulation results to evaluate resonance characteristics, field distributions, and frequency deviations, supporting high-precision microwave system design for space-grade timing applications.

- Collaborated with domain experts to validate simulation methodologies, document technical findings, and contribute to the design optimization of precision frequency standards used in satellite communication and navigation systems.

PROJECTS

FMCG Deal Radar

Tech Stack: Next.js, React, TypeScript, Tailwind CSS, Groq LLM, Hugging Face Embeddings, Vercel Blob, Vercel Cron

- Architected and developed an end-to-end intelligence platform that aggregates, processes, and ranks FMCG mergers, acquisitions, funding, and investment news through an eight-stage agentic data pipeline, transforming unstructured data into actionable business insights.
- Engineered a scalable information processing pipeline comprising ingestion, data cleaning, semantic de-duplication, relevance classification, structured information extraction, credibility scoring, ranking, and AI-generated newsletter creation using Groq LLMs.
- Implemented semantic de-duplication using **all-MiniLM-L6-v2** embeddings with cosine similarity matching, calibrating thresholds across **7,600+** article pairs to maximize duplicate detection while minimizing false merges.
- Optimized inference cost and pipeline efficiency by designing a three-stage relevance filtering strategy (Regex → LLM Screening → LLM Confirmation), reducing LLM token consumption by approximately **75%** while maintaining high recall for relevant business events.
- Built production-ready export services supporting **DOCX, XLSX, PPTX, CSV, and JSON** formats, along with automated snapshot persistence using **Vercel Blob** and scheduled refresh workflows through **Vercel Cron**, enabling reliable data distribution and reporting.

NLP-Based Movie Recommendation System

Tech Stack: Next.js, FastAPI, LangChain, Hugging Face Transformers, FAISS, PostgreSQL, Firebase Authentication, Docker

- Designed and developed a full-stack movie recommendation platform leveraging semantic search and vector embeddings to deliver personalized recommendations based on movie plots, genres, keywords, and user preferences.
- Built scalable FastAPI backend services to generate and index embeddings for **5,000+** movie records using LangChain and Hugging Face models, enabling low-latency similarity search through **FAISS** vector indexing.
- Developed a responsive Next.js frontend with secure Firebase Authentication and PostgreSQL integration, providing seamless user management, personalized recommendations, and efficient metadata retrieval.
- Optimized retrieval performance through vector indexing, embedding optimization, and efficient API design, achieving **sub-second** recommendation latency while ensuring scalable and maintainable deployment using Docker.

Personal Expense Tracker

Tech Stack: Next.js, React, TypeScript, PostgreSQL, Prisma ORM, Tailwind CSS

- Designed and developed a full-stack expense management application enabling users to securely track income, expenses, budgets, and financial transactions through an intuitive dashboard with responsive user experience.
- Engineered scalable REST APIs and a normalized PostgreSQL database schema using Prisma ORM to support transaction management, category-based organization, budgeting, and efficient financial data retrieval.
- Built interactive analytics dashboards featuring expense categorization, monthly spending trends, and visual reports, providing users with actionable insights into their financial habits through dynamic charts and summaries.
- Implemented secure authentication, form validation, reusable UI components, and optimized database queries to deliver a maintainable, scalable, and production-ready application following modern full-stack development practices.

Brain MRI Tumor Classification Using CNN

Tech Stack: Python, TensorFlow, Keras, Scikit-learn, OpenCV, Matplotlib

- Designed and trained a custom Convolutional Neural Network (CNN) for automated brain tumor classification, incorporating image preprocessing, normalization, and data augmentation techniques to improve model robustness and generalization.
- Optimized model performance through architecture refinement, hyperparameter tuning, and regularization strategies, improving classification accuracy by 12% over the baseline while achieving a 94% F1-score on validation data.
- Built a complete machine learning pipeline covering data preprocessing, model training, evaluation, and inference, ensuring reproducible experimentation and efficient model development.
- Evaluated model performance using precision, recall, F1-score, confusion matrix, and ROC analysis to assess classification reliability and support informed model selection.

EDUCATION

Delhi Technological University (DTU) — Oct 2022 – May 2026
- Bachelor of Technology (B.Tech.) in Engineering Physics

Ryan International School, Vasant Kunj — Apr 2021 – Mar 2022
- Senior Secondary Education (CBSE – Class XII)

Ryan International School, Vasant Kunj — Apr 2019 – Mar 2020
- Secondary Education (CBSE – Class X)

CERTIFICATES

- Google Data Analytics Professional Certificate – View Credential
- IBM AI Enterprise Workflow Specialization – View Credential

ACHIEVEMENTS

- Published research paper **"Innovative Cavity Design for Hydrogen Masers Using Simulation Methods"** in ISJEM (March 2026).